

Graphing Quadratics from Vertex or Standard Form

Answer Key

Algebra 1 Parabola Features and Transformations

Quick Answers

- | | |
|--------------|--------------------------|
| 1. -1 | 7. -3, 3 |
| 2. 4 | 8. -2, -4, 0 |
| 3. -4, 5 | 9. -4, -3, 1 |
| 4. -1, 5 | 10. 1, 3, -2 |
| 5. -8, -1, 3 | 11. $(x - 3)^2 - 5$, -5 |
| 6. 9, 5 | 12. 0 |
-

Common wrong answers

1. If they got 15: read the sign of h straight off the parentheses, giving a vertex x of -2 and a y -value of 15
3. If they got 32: used $x = b/2a$ instead of $-b/2a$, landing on $x = -3$
6. If they got -7: dropped the minus sign on a when computing $-b/2a$, landing on $x = -2$
-

1. $y = (x - 2)^2 - 1$.

Step 1. This is vertex form $y = a(x - h)^2 + k$ with $a = 1$, $h = 2$, $k = -1$. The subtraction inside the parentheses is what makes h positive.

Step 2. Check by substituting the axis value: $y(2) = (2 - 2)^2 - 1 = -1$, which is the lowest point because $a = 1 > 0$.

Step 3. Vertex $\boxed{(2, -1)}$, opening upward.

Sketch. Axis $x = 2$; y -intercept $y(0) = 3$; by symmetry the curve also passes through $(4, 3)$, and the x -intercepts are 1 and 3.

Common wrong answer: a vertex y -value of 15 — the sign of h was read straight off the parentheses, putting the vertex at $x = -2$.

2. $y = -(x + 3)^2 + 4$.

Step 1. Rewrite the inside as $(x - (-3))$, so $h = -3$ and $k = 4$. The leading coefficient is -1 .

Step 2. $y(-3) = -(0)^2 + 4 = 4$, and because $a = -1 < 0$ this point is a maximum: the parabola opens downward.

Step 3. Vertex $\boxed{(-3, 4)}$, opening downward.

Sketch. Axis $x = -3$; y -intercept $y(0) = -9 + 4 = -5$; symmetric point $(-6, -5)$; x -intercepts at -5 and -1 .

3. $y = x^2 - 6x + 5.$

Step 1. Standard form with $a = 1$, $b = -6$, $c = 5$. The axis of symmetry is $x = -\frac{b}{2a} = -\frac{-6}{2} = 3.$

Step 2. Substitute to get the vertex y -value: $y(3) = 9 - 18 + 5 = -4.$

Step 3. Vertex $(3, -4)$; the y -intercept is c , that is $y(0) = 5$

Sketch. Opens up; x -intercepts from $(x - 1)(x - 5)$ are 1 and 5, and they straddle $x = 3$ as they must.

Common wrong answer: 32 — $x = \frac{b}{2a}$ was used instead of $-\frac{b}{2a}$, putting the axis at $x = -3.$

4. $y = x^2 - 4x - 5.$

Step 1. The x -intercepts are where $y = 0$, so factor: two numbers multiplying to -5 and adding to -4 are -5 and $1.$

Step 2. $x^2 - 4x - 5 = (x - 5)(x + 1)$, so $x - 5 = 0$ or $x + 1 = 0.$

Step 3. x -intercepts $x = -1$ and $x = 5$

Sketch. The axis of symmetry is midway between them, at $x = 2$, and $y(2) = -9$, so the vertex is $(2, -9).$ The y -intercept is $-5.$

5. $y = 2(x - 1)^2 - 8.$

Step 1. Vertex form with $h = 1$, $k = -8$, $a = 2$, so $y(1) = -8$ and the parabola opens up, narrower than $y = x^2.$

Step 2. For the x -intercepts set $y = 0$: $2(x - 1)^2 = 8$, so $(x - 1)^2 = 4$ and $x - 1 = \pm 2.$

Step 3. x -intercepts $x = -1$ and $x = 3$

Sketch. Vertex $(1, -8)$, axis $x = 1$, y -intercept $y(0) = -6.$

Note: vertex form hands you the intercepts in two moves — no factoring and no quadratic formula.

6. $y = -x^2 + 4x + 5.$

Step 1. Here $a = -1$, $b = 4$, $c = 5$. Axis of symmetry: $x = -\frac{4}{2(-1)} = 2.$ Keeping the sign of a is the whole difficulty.

Step 2. $y(2) = -4 + 8 + 5 = 9.$ Because $a < 0$, this is a *maximum*.

Step 3. Vertex $(2, 9)$, a maximum; y -intercept $y(0) = 5$

Sketch. Opens down; $-x^2 + 4x + 5 = -(x - 5)(x + 1)$, so the x -intercepts are -1 and $5.$

Common wrong answer: -7 — the minus sign on a was dropped in $-\frac{b}{2a}$, putting the axis at $x = -2.$

7. $y = x^2 - 9.$

Step 1. There is no x term, so $b = 0$ and the axis of symmetry is $x = -\frac{0}{2} = 0$ — the y -axis itself.

Step 2. Set $y = 0$: $x^2 = 9$, a difference of squares $(x - 3)(x + 3) = 0.$

Step 3. x -intercepts $x = -3$ and $x = 3$

Sketch. Vertex $(0, -9)$ on the y -axis, opening up. The two intercepts are mirror images precisely because the axis is $x = 0.$

8. $y = \frac{1}{2}(x + 2)^2 - 2$.

Step 1. Vertex form with $h = -2$, $k = -2$, $a = \frac{1}{2}$. Since $|a| < 1$, the parabola is *wider* than $y = x^2$: at one unit from the axis it rises only $\frac{1}{2}$ instead of 1.

Step 2. $y(-2) = \boxed{-2}$, the minimum. For the intercepts set $y = 0$: $\frac{1}{2}(x + 2)^2 = 2$, so $(x + 2)^2 = 4$ and $x + 2 = \pm 2$.

Step 3. x -intercepts $\boxed{x = -4 \text{ and } x = 0}$

Sketch. Vertex $(-2, -2)$, axis $x = -2$; the curve passes through the origin, which is one of the two intercepts.

9. $y = x^2 + 2x - 3$.

Step 1. $a = 1$, $b = 2$: the axis is $x = -\frac{2}{2} = -1$, and $y(-1) = 1 - 2 - 3 = -4$.

Step 2. Factor for the intercepts: two numbers with product -3 and sum 2 are 3 and -1 , giving $(x + 3)(x - 1)$.

Step 3. Vertex $\boxed{(-1, -4)}$; x -intercepts $\boxed{x = -3 \text{ and } x = 1}$

Sketch. Opens up, y -intercept -3 . Notice the vertex x -value -1 is the average of -3 and 1 — the two routes agree, which is a free check.

10. $y = 2x^2 - 8x + 6$.

Step 1. Factor the leading coefficient out first: $2(x^2 - 4x + 3) = 2(x - 1)(x - 3)$. A common factor left in place is where sign errors breed.

Step 2. Setting each factor to zero gives the x -intercepts $\boxed{x = 1 \text{ and } x = 3}$

Step 3. The axis of symmetry is midway between the intercepts, at $x = 2$, and $y(2) = 8 - 16 + 6 = -2$, so the vertex is at $\boxed{(2, -2)}$

Sketch. Opens up and is narrow ($a = 2$); the y -intercept is 6 .

11. $y = x^2 - 6x + 4$ — **complete the square.**

Step 1. Take half of $b = -6$, giving -3 , and square it: 9 . Add and subtract that inside the expression: $x^2 - 6x + 9 - 9 + 4$.

Step 2. The first three terms are a perfect square: $(x - 3)^2 - 5$. Expanding back gives $x^2 - 6x + 9 - 5 = x^2 - 6x + 4$, so the two forms are the same function.

Step 3. Vertex form: $\boxed{y = (x - 3)^2 - 5}$, so the vertex is $(3, \boxed{-5})$

Sketch. Axis $x = 3$, opens up, y -intercept 4 ; the x -intercepts are $3 \pm \sqrt{5}$, about 0.8 and 5.2 .

12. Vertex $(3, -4)$, through $(5, 0)$. *The sketch and the explanation are open work, flagged for manual review; the model answer below is what full credit contains.*

Step 1. A vertex of $(3, -4)$ fixes h and k , so the equation must look like $y = a(x - 3)^2 - 4$. Only a is still unknown, and one more point is exactly enough to find it.

Step 2. Substitute $(5, 0)$: $0 = a(5 - 3)^2 - 4 = 4a - 4$, so $a = 1$ and the equation is $y = (x - 3)^2 - 4$.

Step 3. Check the given point in the finished equation: $(5 - 3)^2 - 4 = 4 - 4$, so $y = \boxed{0}$ at $x = 5$, as required.

Step 4 (the sketch). Vertex $(3, -4)$, axis $x = 3$, opens up. x -intercepts: $(x - 3)^2 = 4$ gives $x = 1$ and $x = 5$. y -intercept $y(0) = 5$.

Full credit: the equation $y = (x - 3)^2 - 4$, a sketch labelled with the vertex, the axis $x = 3$ and the intercepts 1 and 5 , and a sentence saying that the vertex fixes h and k while the extra point is the only thing that can determine the stretch factor a .